

From the defn. of symmetric difference, $[x \in A \oplus B]$.

By Defn. 3 from Sec-2.1, $[A \cup B] - (A \cap B) \subseteq A \oplus B$.

$$\therefore [A \oplus B = (A \cup B) - (A \cap B)] \square$$

42) Show that $A \oplus B = (A - B) \cup (B - A)$

Solution: $(A - B) \cup (B - A) = \{x \mid x \in (A - B) \vee x \in (B - A)\}$ (Defn. 1)
 $= \{x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$ (Defn. 4)
 $= A \oplus B$ (From the derivation of Problem 41) $\square$

43) Show that if A is a subset of a universal set U , then

a) $A \oplus A = \emptyset$

Solution: $A \oplus A = (A \cup A) - (A \cap A)$ (Prob 41)
 $= A - (A \cap A)$ (1st Identity Law)
 $= A - A$ (2nd Identity Law)
 $= A \cap \bar{A}$ (Prob 21(a))
 $= \emptyset$ (Prob 9(b)) $\square$

b) $A \oplus \emptyset = A$

Solution: $A \oplus \emptyset = (A \cup \emptyset) - (A \cap \emptyset)$ (Prob 42)
 $= A \cup (\emptyset - A)$ (Prob 10(a))
 $= A \cup \emptyset$ (Prob 10(b))
 $= A$ (2nd Identity Law) $\square$

c) $A \oplus U = \bar{A}$

Solution: $A \oplus U = (A \cup U) - (A \cap U)$ (Prob 41)
 $= U - (A \cap U)$ (1st Domination Law)
 $= U - A$ (1st Identity Law)
 $= U \cap \bar{A}$ (Prob 21(a))
 $= \bar{A} \cap U$ (2nd Commutative Law)
 $= \bar{A}$ (1st Identity Law) $\square$

d) $A \oplus \bar{A} = U$

Solution: $A \oplus \bar{A} = (A \cup \bar{A}) - (A \cap \bar{A})$ (Prob 41)
 $= U - (A \cap \bar{A})$ (1st Complement Law)
 $= U - \emptyset$ (2nd Complement Law)
 $= U$ (Prob 10(a)) $\square$